PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	A Qualitative Focus Group Interview Study of Communication between Patients, Dentists and Physicians for Efficient Osteonecrosis of the Jaw Practices
AUTHORS	Chueh, Ting-Yu; Chen, Yung-Chih; Hung , Tsung-Min

VERSION 1 – REVIEW

REVIEWER	Carter, Sophie E York St John University
REVIEW RETURNED	07-Apr-2021

GENERAL COMMENTS	Thank you for the invitation to review this manuscript. The manuscript is well written and considers two important factors, cognitive function and cerebrovascular function, in the context of prolonged sitting with or without physical activity breaks. This is an emerging area, evidenced by the fact the first included paper for CBF was published in 2018 and for cognition 2016. As such, it does not seem appropriate to be systematically reviewing this area, given the small number of studies conducted to date. Indeed, only 3 were included for cerebrovascular function. More studies focusing on sitting and cerebrovascular function have been published since the authors finished their search. Consequently, it does not seem appropriate or timely to conduct a systematic review on such an under-researched field, as there is minimal research to synthesise and the only real conclusions that can be made are that more research is needed. Other comments are listed below: General 1. The authors describe that their final search was completed on the 20th September 2020, yet a study by Maasakkers et al. investigated the short-term effects of three hours uninterrupted sitting with/without PA breaks on cognitive and cerebrovascular outcomes (doi: 10.1186/s13195-020-00644-z), which was published in June 2020, is not included. 2. The authors aim (Number 2) was to provide a methodological assessment of all included studies, however key factors related to methods employed in the studies, such as differences in the assessment of cognitive function and cerebrovascular function, and limitations surrounding approaches used, are not discussed. Introduction: 3. Lines 5-6: ‘constant cerebral perfusion of cerebral blood flow (CBF) to provide sufficient energy..’ This does not make clear sense, please revise

	4. Lines 16-17: It would be beneficial to mention here the evidence that SB is an independent risk factor from PA/exercise 5. Lines 18-19: This sentence does not seem appropriate. Even if someone is overweight structured exercise may be advised as a weight loss intervention and to confer other health benefits 6. Lines 19-21: 'Accordingly, it would be very useful to understand the optimal mix of physical activity (PA) and sedentary behavior in the course of each day.' Optimal mix of what? This statement it unclear. Also, considering for other health outcomes that are much more researched in relation to prolonged sitting, such as metabolic markers, the 'optimal mix of PA and SB is not clear, this seems an advanced statement in the context of CBF and cognition. Indeed, the recently published WHO guidelines stated there is insufficient evidence to provide quantitative SB guidelines. 7. Lines 22-24: It would be beneficial to briefly outline existing research in this area that has shown beneficial effects adopting this strategy on other physiological outcomes such as metabolic makers and vascular function. 8. Line 23: Less focus should be given on the practicalities of frequency PA breaks, since this is not the focus of the paper. Instead, the rationale should be based on existing literature showing the patterning over which sitting is accrued during the day can have negative consequences to health 9. Line 25: Studies have also explored the impact of taking sitting breaks on cerebrovascular function 10. Lines 27-28: 'understanding the physiological mechanisms behind the acute effect of taking PA breaks from sitting is important.' This statement needs to be more specific in line with the focus of the paper 11. Line 29: This should be 'aims' as the authors then proceed to list three aims. Methods 12. Lines 60-61: The authors should also include terms such as 'cerebrovascular function' (especially as this term is used frequently in the introduction of the manuscript) or 'continuous sitting' in their search strategy 13. Lines 60-61: The authors should also specify where the search was conducted e.g., did the search journal titles, or titles and abstract? Discussion 14. Section- The acute effects of breaking sitting on cognitive performances: Some mention must be given to the fact that studies have assess different cognitive outcomes (and likely used different cognitive tests to assess similar outcomes). The authors discuss differences in break strategy influencing cognitive outcomes, yet the cognitive outcomes assessed (e.g., WM, inhibitory control) vary between these studies, in addition to the break strategy 15. Lines 218-220: The authors should state here how long each PA break was, in addition to the accumulative exercise total 16. Limitations: The authors need to discuss the heterogeneity in assessments/testing batteries used to assess cognitive function. Also, a limitation is the acute nature of studies, and future research should explore the accumulative effects of multiple days of sitting with or without PA breaks. With regards CBF, the authors need to highlight methodological limitations associated with the assessment of just MCAv or specific cortical regions.

REVIEWER	Gordon, Carron University of the West Indies
REVIEW RETURNED	14-Jun-2021

GENERAL COMMENTS	1. A few minor errors should be corrected. Page 1, line 16; page 3 lines 57 and 58; page 4 line 82; page 7 line 136; page 10 line 208; page 12 line 256; page 13 lines 276 and 284. 2. Abstract is incomplete. Some items from PRISMA 2020 Abstract Checklist are missing. For example,number of participants in each study, primary source of funding for review.

REVIEWER	Hoffmann, Coles Arizona State University
REVIEW RETURNED	21-Sep-2021

GENERAL COMMENTS	Abstract: It is suggested that the authors include information like total sample size, and age range of sample. Given the small number (3) of the studies and the lack of overlap in outcomes between the studies, it is suggested that the cerebrovascular function is excluded from the review. Introduction: (Line 2) It is suggested that the importance of optimal cognitive function is included as a leading sentence in the introduction (e.g. why it is important to optimize daily behaviors, reduce cognitive declime). In the first paragraph I think the relationship between cognitive function and cerebrovascular function should be elaborated on, to establish a foundation for why you have included both outcomes. Information and search strategy: What were the inclusion and exclusion criteria? Synthesis of Results The results in this review were analyzed (remove “by”) through… Characteristics of Included Studies (Line 115) Instead of “Most of the participants (54%)” I suggest saying “More than half” Cognitive function measurements and results I suggest clarifying the results- when you summarize each point are you talking about results from each of the studies or overarching results (and if so, how many of the studies reported each of these. For example, you could say something like “Two of the seven studies reported significant improvements in executive function.” The acute effects of breaking sitting on cognitive performance: First sentence may be misleading considering the limited evidence. Perhaps be more cautiously optimistic and then contrast the following 2 sentences (In study … in contrast…) (Line 206) I suggest adding a sentence before the example of the meta-analysis to explain that the results may have been impacted by the intensity of the PA interventions. The acute effects of breaking sitting on cerebrovascular responses: Do any of the studies that measure the CBF also measure cognitive function? It is suggested that the authors consider removing this result from the review and perhaps mention the evidence that they can be related (as a potential mechanism). Limitations: Several limitations should be mentioned for example small sample size, heterogeneity (of study, of sample), different measurements of cognitive function. Conclusion: It is suggested that the results really be interpreted with caution, this is a very important topic and undoubtedly an area that deserves further exploration, but it should be emphasized that less than half of the studies had significant

	results (it may be helpful to elaborate on the studies that didn't have significant effects— did they have smaller samples? Shorter trials? Different types of PA? Etc.) It would be helpful to include more information on the studies with significant outcomes for intervention group - RCT? Sample size? PA duration, intensity, length, type? in the results section (you followed up well with this information in the discussion) in contrast to studies without significant outcomes. I think it also worth discussing whether the improvements in in the outcomes could be due to breaking up sedentary time or just possibly attributable to PA (regardless of sedentary time) (as previous studies have asserted)

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1 Dr. Sophie E Carter

Comments to the author	Response to reviewer
Thank you for the invitation to review this manuscript. The manuscript is well written and considers two important factors, cognitive function and cerebrovascular function, in the context of prolonged sitting with or without physical activity breaks. This is an emerging area, evidenced by the fact the first included paper for CBF was published in 2018 and for cognition 2016. As such, it does not seem appropriate to be systematically reviewing this area, given the small number of studies conducted to date. Indeed, only 3 were included for cerebrovascular function. More studies focusing on sitting and cerebrovascular function have been published since the authors finished their search. Consequently, it does not seem appropriate or timely to conduct a systematic review on such an under-researched field, as there is minimal research to synthesise and the only real conclusions that can be made are that more research is needed. Other comments are listed below:	Thank you for taking the time to provide professional suggestions and comments in this manuscript. We believe the manuscript has been improved after taking your comments into account. We respected the reviewer's concerns about the appropriateness of conducting a systematic review on an emerging topic with only a few empirical studies included. Given that breaking sitting and brain health is critical issues, we argued that it is necessary to systematically review the related studies for providing current evidence and suggestions for future studies. Moreover, several published systematic review studies were also with six to seven studies (Biber & Ellis, 2019; Carson et al., 2016; Sibandze & Scafide, 2018; Xiong et al., 2015). Nevertheless, we understood your and the other reviewer's concern that only 3 cerebrovascular function studies were included in this review study. Thus, we decided to exclusively focus on studies targeting cognition in this systematic review. To clearly address your other comments, we have made point-by-point replies in the following columns and highlighted YELLOW color for our revisions in this letter and the manuscript. As some comments were the same as other reviewers, we highlighted GRAY color in response to these comments. We hope our replies and revisions can meet with your approval.
General 1. The authors describe that their final search was completed on the 20th September 2020, yet a study by Maasakkers et al. investigated the short-term effects of three hours uninterrupted sitting with/without PA breaks on cognitive and cerebrovascular outcomes	Thank you for the reminder. The present study exclusively focused on healthy participants (please see lines 47–49). Although the study you mentioned (Maasakkers et al., 2020) appeared in the results of our early search stage, we decided to exclude it due to that some participants had mild cognitive impairments (MCI) based on scores of

(doi: 10.1186/s13195-020-00644-z), which was published in June 2020, is not included.	Montreal Cognitive Assessment i.e., < 26 (Nasreddine et al., 2005).
2.The authors aim (Number 2) was to provide a methodological assessment of all included studies, however key factors related to methods employed in the studies, such as differences in the assessment of cognitive function and cerebrovascular function, and limitations surrounding approaches used, are not discussed.	Thank you for these helpful suggestions. We have covered these issues in the discussion section. Please see the following reply (No. 16).
Introduction: 3. Lines 5-6: 'constant cerebral perfusion of cerebral blood flow (CBF) to provide sufficient energy..' This does not make clear sense, please revise	As we decided to exclude the cerebrovascular function studies in this review based on your and other reviewers' comments, we removed this sentence.
4.Lines 16-17: It would be beneficial to mention here the evidence that SB is an independent risk factor from PA/exercise.	Thank you for this suggestion. We added a sentence to mention that the prolonged sitting was an independent risk factor of cognitive performance from PA. Please see our revision below: Page 1, line 17: This association was partially independent of the PA level (Ku et al., 2017).
5.Lines 18-19: This sentence does not seem appropriate. Even if someone is overweight structured exercise may be advised as a weight loss intervention and to confer other health benefits.	We respected the reviewer's comment about this point. Thus, we removed this sentence.

6. Lines 19-21: 'Accordingly, it would be very useful to understand the optimal mix of physical activity (PA) and sedentary behavior in the course of each day.' Optimal mix of what? This statement it unclear. Also, considering for other health outcomes that are much more researched in relation to prolonged sitting, such as metabolic markers, the 'optimal mix of PA and SB is not clear, this seems an advanced statement in the context of CBF and cognition. Indeed, the recently published WHO guidelines stated there is insufficient evidence to provide quantitative SB guidelines.	Thank you for this comment. We revised this sentence as below: Page 1, lines 22 to 23: Accordingly, taking regular PA breaks from sitting throughout the day may be a feasible strategy to offset the negative effects of prolonged sitting on cognition.
7. Lines 22-24: It would be beneficial to briefly outline existing research in this area that has shown beneficial effects adopting this strategy on other physiological outcomes such as metabolic makers and vascular function.	Thank you for this helpful suggestion. We have briefly outlined the results related to breaking sitting and physiological outcomes such as metabolic makers and vascular function in the opening of this paragraph. Page 1, lines 24 to 25: Taking regular PA breaks from sitting has been shown to be a practical approach to reduce the negative effects of acute prolonged sitting on cardiovascular and metabolic functions (Chen et al., 2018; Dunstan et al., 2012; Saunders et al., 2018)
8. Line 23: Less focus should be given on the practicalities of frequency PA breaks, since this is not the focus of the paper. Instead, the rationale should be based on existing literature showing the patterning over which sitting is accrued during the day can have negative consequences to health.	Thank you for this suggestion. We revised the rational of this study as below: Page 2, lines 26 to 28: Given that acute uninterrupted sitting has negative effects on cognition (Baker et al., 2018; Triglav et al., 2019), related studies have extended the investigation from the physiological outcomes to the cognitive outcomes of taking regular breaks from sitting.
9. Line 25: Studies have also explored the impact of taking sitting breaks on cerebrovascular function	As we decided to exclusively focus on studies targeting cognition in this systematic review, we did not added this sentence.

10.Lines 27-28: ‘understanding the physiological mechanisms behind the acute effect of taking PA breaks from sitting is important.’ This statement needs to be more specific in line with the focus of the paper	Thank you for this suggestion. We have added the primary outcome of this study in this sentence for clarity. Please see our revision below: Page 2, line 34: Furthermore, understanding the physiological mechanisms behind the acute effects of taking PA breaks from sitting on cognition is important to facilitate the design of successful PA approaches.
11.Line 29: This should be ‘aims’ as the authors then proceed to list three aims.	Thank you for pointing out this grammar issue. We have corrected it accordingly. Page 2, line 33: Thus, the aims of this paper were (1) to……
Methods 12. Lines 60-61: The authors should also include terms such as ‘cerebrovascular function’ (especially as this term is used frequently in the introduction of the manuscript) or ‘continuous sitting’ in their search strategy.	Thank you for this suggestion. We have added the term–continuous sitting in the search keywords and revised some texts to clarify our search method . Please our revision below: Page 3, line 64: Key words were as follows: ("Prolonged sitting" OR "Interrupted sitting" OR "breaking sitting" OR "uninterrupted sitting" OR " continuous sitting ") AND (cognitive OR cognition OR executive OR brain OR cerebral).
13. Lines 60-61: The authors should also specify where the search was conducted e.g., did the search journal titles, or titles and abstract?	Thank you for the reminder. We added some texts to clarify our search method . Please see our revision below: Page 3, line 61 to 62: The key words used to search the titles and abstracts were discussed by the research team to maximize the chance of identifying relevant articles
Discussion 14. Section- The acute effects of breaking sitting on cognitive performances: Some mention must be given to the fact that studies have assess different cognitive outcomes	Thank you for this suggestion. We added some texts to mention this issue in the limitation section. Please see our following reply (No.16).

(and likely used different cognitive tests to assess similar outcomes). The authors discuss differences in break strategy influencing cognitive outcomes, yet the cognitive outcomes assessed (e.g., WM, inhibitory control) vary between these studies, in addition to the break strategy.	
15. Lines 218-220: The authors should state here how long each PA break was, in addition to the accumulative exercise total	Thank you for this suggestion. We have added the information related to duration of PA breaks in this study (Mullane et al., 2017). Please see our revision below: Page 9, lines 210 to 211: Interestingly, taking hourly low-intensity PA breaks might still benefit cognition if these breaks are sufficiently long (i.e., 10, 15, 20, and 30 min).
16. Limitations: The authors need to discuss the heterogeneity in assessments/testing batteries used to assess cognitive function. Also, a limitation is the acute nature of studies, and future research should explore the accumulative effects of multiple days of sitting with or without PA breaks. With regards CBF, the authors need to highlight methodological limitations associated with the assessment of just MCAv or specific cortical regions.	Thank you for these suggestions. We have added some texts to discuss these issues. As we decided to exclude the cerebrovascular function studies in this review, there was no further discussion on this outcome. Please see our revisions in response to your first two comments below: Page 13, lines 323 to 330: Second, given the heterogeneity of PA protocols, cognitive assessments, and sample size across the included studies, more studies with larger samples are needed to clarify the role of different PA characteristics (i.e., frequency, intensity, duration, and type) in the effect of PA breaks from sitting on distinct cognitive domains so as to provide more precise recommendations for public health. Third, this review exclusively focused on studies that assessed the acute effects of PA breaks on cognition. Future research should explore the accumulative effects of multiple days of sitting with or without PA breaks on cognition.

Reviewer: 2 Dr. Carron Gordon

Comments to the Author:	Response to reviewer
1. A few minor errors should be corrected. Page 1, line 16; page 3 lines 57 and 58; page 4 line 82; page 7 line 136; page 10	Thank you for pointing out these errors. We have corrected these grammar issues and typo accordingly. The revised manuscript has been

line 208; page 12 line 256; page 13 lines 276 and 284.	proofread by native speaker to avoid grammar issues before the resubmission. Please see our corrections below: Page 1, line 16 (now is in line 20): cognitive performance is enhanced after 20 to 30 min of continuous/structured exercise Page 3, lines57 (now is in line 60): initial search was conducted on the… Page 3, line 58 (now is in line 61): the final search… Page 4 line 82 (now is in line 84): The data extracted from the studies were analyzed using a narrative synthesis approach Page 7 line 136: we removed this sentence because we decided to exclusively focus on studies targeting cognition in this systematic review. Page 10, line 208 (now is in lines 200 to 201): but the effect diminishes when the exercise intensity is very light. Page 12, line 256 (now is in line 248): Of note, one study found adverse effects of frequent calf raise exercises on inhibitory control in young adults Page 13, line 276 and 284: we removed these sentence that related to cerebrovascular function because we decided to exclusively focus on studies targeting cognition in this systematic review based on other reviewer's comment.
2. Abstract is incomplete. Some items from PRISMA 2020 Abstract Checklist are missing. For example, number of participants in each study, primary source of funding for review.	Thank you for the reminder. We have added the necessary information such as number of participants, primary source of funding to comply with the PRISMA 2020 Abstract Checklist. Please see our revisions below: Page 2, line 42: Results: Seven studies, involving 168 participants aged between 18 and 80 years, were eligible for inclusion in this review. Page 2, lines 55: Funding: This study was financially supported by the Ministry of Education.

Reviewer: 3 Dr. Coles Hoffmann

Comments to the Author:	Response to reviewer
1.This review covers a very important research area that is in need of attention. The review examines whether breaking up sedentary time with physical activity impacts cognitive function and cerebrovascular function. This is an area of research that could greatly inform future interventions and public health messaging to prevent cognitive decline.	Thank you for taking the time to provide helpful suggestions and comments in this manuscript. We believe the manuscript has been improved after taking your comments into account. To clearly address your comments, we have made point-by-point replies in the following columns and highlighted BLUE color for our revisions in this letter and the manuscript. As some comments were the same as other reviewers, we highlighted GRAY color in response to these comments. We hope our replies and revisions can meet with your approval.
2.Abstract: It is suggested that the authors include information like total sample size, and age range of sample. Given the small number (3) of the studies and the lack of overlap in outcomes between the studies, it is suggested that the cerebrovascular function is excluded from the review.	Thank you for these suggestions. We have added some information in the abstract. In addition, we accepted your suggestion to exclude cerebrovascular function from the current review. Please see our revision below: Page 1, line 42: Results: Seven studies, involving 168 participants aged between 18 and 80 years, were eligible for inclusion in this review.
3.Introduction: (Line 2) It is suggested that the importance of optimal cognitive function is included as a leading sentence in the introduction (e.g. why it is important to optimize daily behaviors, reduce cognitive decline). In the first paragraph I think the relationship between cognitive function and cerebrovascular function should be elaborated on, to establish a foundation for why you have included both outcomes.	Thank you for these suggestions. First, we highlighted the importance of cognition in the opening sentence in the first paragraph and added some texts for clarifying its importance. Second, as we decided to exclude the cerebrovascular function studies in this review, we removed texts related to this issue in this paragraph. Please see our revisions below: Page 1, lines 2 to 11: Optimal cognitive performance in all cognitive domains, including attention, executive functions (EFs), and memory engagement, is important for optimizing daily functioning and reducing cognitive decline, which are critical to productivity and quality of life (Hill et al., 2017; Sanz et al., 2018). In classrooms and offices, for example, individuals require attention to focus on relevant information while relying on distinct EFs to suppress irrelevant distraction (inhibitory control), hold and mentally organize information (working memory [WM]), and adjust their behavior or thoughts based on updated demands, rules, or priorities (cognitive flexibility) to achieve goals (Diamond, 2013; Miller & Cohen, 2001; Miyake et al., 2000; Zelazo, 2015). These abilities have profound attribution effects on academic attainment and job performance (Lang et

	al., 2010; Rohde & Thompson, 2007). In addition, a healthy cognitive state is beneficial in late life, because it reduces age-related cognitive decline (Buckner, 2004). Therefore, it is important to maintain and/or enhance cognition in daily life.
4.Information and search strategy: What were the inclusion and exclusion criteria?	Please see the inclusion and exclusion criteria in the Eligibility criteria section. Pages 2 to 3, lines 46 to 56: The Population, Intervention, Comparison, Outcome, and Study (PICOS) framework was used to determine the inclusion criteria for studies (Moher et al., 2015), as follows. (P) Participants: participants were healthy and without any cardiovascular disease, history of brain injury, or psychiatric or neurological disorder. (I) Intervention: interventions were of an acute nature, involving interrupting prolonged periods of sitting with a regular pattern of PA at predefined intervals (e.g., every 30 min). (C) Comparator: studies included a non-treatment control group (i.e., the prolonged sitting condition). (O) Outcome: studies tested cognitive performance. (S) Study design: studies were randomized controlled or non-randomized controlled trials. Studies were excluded if they (1) involved only a single bout of PA in the intervention condition; (2) did not explicitly state the PA-related parameters, such as type, intensity, frequency, and duration; and/or (3) were not published in an English language peer-reviewed journal.
5.Synthesis of Results The results in this review were analyzed (remove “by”) through...	Thank you for the reminder. We have corrected this typo. Please see our correction below: Page 4, line 84: The results in this review were analyzed through a process of narrative synthesis
Characteristics of Included Studies (Line 115) Instead of “Most of the participants (54%)” I suggest saying “More than half”	Thank you for the suggestion. We have rephrased this sentence accordingly. Please see our revision below: Page 5, lines 117 to 118:: More than half participants (57%) were overweight or obese (BMI > 25 kg/m^2)

6.Cognitive function measurements and results I suggest clarifying the results- when you summarize each point are you talking about results from each of the studies or overarching results (and if so, how many of the studies reported each of these. For example, you could say something like “Two of the seven studies reported significant improvements in executive function.”	Thank you for this suggestion. We have mentioned the overarching results before we summarized results from each of the studies. Please see our revisions below: Page 6, lines142 to 143: Three of the seven studies found the positive effects of breaking sitting on EFs.
7.The acute effects of breaking sitting on cognitive performance: First sentence may be misleading considering the limited evidence. Perhaps be more cautiously optimistic and then contrast the following 2 sentences (In study ... in contrast...)	Thank you for the comment. We have revised the first sentence accordingly. Please see our revision below: Page 8, lines 191 to 192: This review suggests that taking PA breaks from sitting, specifically their intensity and/or volume, might play a potential role in regulation of cognitive performance.
8.(Line 206) I suggest adding a sentence before the example of the meta-analysis to explain that the results may have been impacted by the intensity of the PA interventions.	Thank you for this suggestion. We have added some information before the example of the meta-analysis to explain that the results may have been impacted by the intensity of the PA interventions. Please see our revision below: Page 8, lines 197 to 198: Studies have reported that exercise intensity may be an important determinant of cognitive benefits (Bailey et al., 2021; Zimmer et al., 2016)
9.The acute effects of breaking sitting on cerebrovascular responses: Do any of the studies that measure the CBF also measure cognitive function? It is suggested that the authors consider removing this result from the review and perhaps mention the evidence that they can be related (as a potential mechanism).	Thank you for this suggestion. We excluded cerebrovascular function studies in this review based on your previous suggestion. One study concurrently measure the cognition and cerebrovascular function (Stoner et al., 2019). We accepted your suggestion to mention cerebrovascular function maybe a potential mechanism of the effects of breaking sitting on cognition in this study. Please see our revision below: Page 10, lines 248 to 252:

	Of note, one study found adverse effects of frequent calf raise exercises on inhibitory control in young adults (Stoner et al., 2019). The impaired inhibitory control performance was accompanied by a decreased prefrontal cortex perfusion, as measured by near-infrared spectroscopy, supporting that prefrontal cortex activity could be a potential mechanism for modulating cognitive performance (Miller & Cohen, 2001).
10.Limitations: Several limitations should be mentioned for example small sample size, heterogeneity (of study, of sample), different measurements of cognitive function.	Thank you for these suggestions. We added some arguments to discuss these issues. Please see the revision below: Page 13, lines 323 to 327: Second, given the heterogeneity of PA protocols, cognitive assessments, and sample size across the included studies, more studies with larger samples are needed to clarify the role of different PA characteristics (i.e., frequency, intensity, duration, and type) in the effect of PA breaks from sitting on distinct cognitive domains so as to provide more precise recommendations for public health.
11.Conclusion: It is suggested that the results really be interpreted with caution, this is a very important topic and undoubtedly an area that deserves further exploration, but it should be emphasized that less than half of the studies had significant results (it may be helpful to elaborate on the studies that didn't have significant effects— did they have smaller samples? Shorter trials? Different types of PA? Etc.). It would be helpful to include more information on the studies with significant outcomes for intervention group - RCT? Sample size? PA duration, intensity, length, type? in the results section (you followed up well with this information in the discussion) in contrast to studies without significant outcomes. I think it also worth discussing whether the improvements in the outcomes could be due to breaking up sedentary time or just possibly attributable to PA (regardless of sedentary time) (as previous studies have asserted)	Thank you for these suggestions. We mentioned that less than half of the studies had significantly positive effects of breaking sitting on cognition in the discussion section and cautiously made the conclusion in this study. In addition, we have added the necessary information on studies with significant outcomes for intervention group in the result section (please see page 6, lines 144 to 154). We also added an additional suggestion for future studies based on your last argument. Please see our revisions in response to your these comments below: Page 2m lines 51 to 54 (abstract section): Given the limited evidence on this topic in the literature and the heterogeneity of PA protocols across the included studies, our results regarding the effectiveness of interrupting prolonged sitting with PA breaks in improving cognition warrant further verification. Page 11, lines 265 to 267:

	Furthermore, the findings of the effectiveness of PA breaks from sitting in improving cognition need to be further confirmed, because fewer than half of the studies included reported significant positive results. Page 16, lines 336 to 342: Conclusions: The current review provides preliminary evidence that the intensity and/or volume of PA breaks from prolonged sitting may determine the effects of such breaks on cognition. However, due to the heterogeneity of PA protocols across studies and limited evidence, firm conclusions regarding the effectiveness of interrupting sitting in improving cognition cannot be drawn. Further studies are warranted to examine the duration, frequency, and intensity of PA breaks required to optimize cognitive performance. Page 14, lines 332 to 334: Finally, future studies should examine the isolated contributions of interrupting prolonged sitting versus continuous PA to improvements in cognition.

VERSION 2 – REVIEW

REVIEWER	Carter, Sophie E York St John University
REVIEW RETURNED	01-Dec-2021

GENERAL COMMENTS	Thank you to the authors for taking the time to read the reviewer comments and amend their manuscript accordingly. The manuscript has been strengthened by the removal of studies assessing cerebrovascular function and the sole focus on studies investigating breaking up sitting on cognition. I have only two minor comments detailed below: Introduction 1. Page 1, Line 10: Please revise to ‘later life’ Discussion 2. Page 12, Lines 301-303: A reference to this study that explored CBF is required

REVIEWER	Hoffmann, Coles Arizona State University
REVIEW RETURNED	14-Jan-2022

GENERAL COMMENTS	Thank you so much for addressing the previous concerns. This is a very important area and a very necessary review. My main suggestion is that the conclusions be updated because I think currently they are misrepresenting the results (3 out of 7 reported positive effects) which is less than half, but the conclusion in the manuscript states that the manuscript provides preliminary evidence that interrupting prolonged sitting may beneficially affect cognition in healthy adults. I think more accurately would be that it is currently unclear and larger RCTs are needed to understand the relationship, especially due to three studies that did find a positive effect.

VERSION 2 – AUTHOR RESPONSE

Reviewer 1: Dr. Sophie E Carter

Comments to the Reviewer 1:	Response to Reviewer 1
Page 1, Line 10: Please revise to 'later life' Discussion	Thank you for this suggestion. We revised this sentence accordingly as below: Page 1, line 10: In addition, a healthy cognitive state is beneficial in later life, because it reduces age-related cognitive decline.
Page 12, Lines 301-303: A reference to this study that explored CBF is required	Thank you for the reminder. We have added a reference accordingly. "Carter, S. E., Draijer, R., Holder, S. M., Brown, L., Thijssen, D. H., & Hopkins, N. D. (2018). Regular walking breaks prevent the decline in cerebral blood flow associated with prolonged sitting. Journal of Applied Physiology, 125(3), 790-798." Please see our revision below: Page 12, line 299 to 301: One study found that in healthy, normal-weight adults, taking short but frequent PA breaks (every 30 min) resulted in higher CBF than taking longer but less frequent PA breaks (every 120 min)[50]

Reviewer 3: Dr. Coles Hoffmann

Comments to the Reviewer 3	Response to Reviewer 3
Thank you so much for addressing the previous concerns. This is a very important area and a very necessary review. My main suggestion is that the conclusions be updated because I think currently they are misrepresenting the results (3 out of 7 reported positive effects) which is less than half, but the conclusion in the manuscript states that the manuscript provides preliminary evidence that interrupting prolonged sitting may beneficially affect cognition in healthy adults. I think more accurately would be that it is currently unclear and larger RCTs are needed to understand the relationship, especially due to three studies that did find a positive effect.	Thank you for this comment. We revised the texts related to conclusions across manuscript based on your suggestion. Please see our revisions below: Abstract-conclusion: Page 2, lines 50 to 52: Given the limited evidence with mixed findings on this topic in the literature and the heterogeneity of PA protocols across the included studies, the results regarding the effectiveness of interrupting prolonged sitting with PA breaks in improving cognition warrant further verification. Discussion: Page 8, lines 182 to 184: Three of the seven studies suggested that taking regular PA breaks from sitting offers short-term benefits for cognition in healthy adults.